Light Painting Classroom

by John Hartman

Photography Tips

Photoshop Tips

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Preparing for the Photo Shoot: Checklist

1. Charge all batteries—camera, lights, triggers, smart phones/tablets/laptops, etc.
2. Format memory card and have a spare formatted card available
3. Camera with lens, tripod
4. Trigger device, viewing device
5. All lights, diffusers, spare batteries
6. Stand for laptop or tablet
7. Chair for person triggering the shutter

Get Your Camera ready

1. f/16 or f/22 aperture
2. ISO 100
3. Shutter set to Bulb setting
4. Manual focus
5. Manual exposure
5. White balance set to between 4000º and 5000ºk (for viewing initial images with reasonable accuracy)
6. Turn on wi-fi on camera and connect to tablet or smart phone (if available)

OR

6. Connect camera to computer and begin tethered capture to your RAW processor
7. Make sure subject is properly focused. You can use your camera's autofocus initially, but once focused turn on manual focusing. If you focus manually, confirm with Live View (if available) or examine test shots zoomed up on your laptop, computer or smartphone.
8. Make and evaluate test exposures before you begin. Once you start light painting, the camera can't be touched for any reason!

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Light Painting Steps–Tabletop Still Life

1. Once composition, view angle and focal length are determined, finalize focus using manual focus mode
2. Place background several feet behind subject and cover background with black material
3. Create Base Layer file moving bright light behind, above and to the sides of the camera
4. Be sure to take note of which lights you use on each shot (for future white balancing)
5. Light paint individual objects first with small flashlight and 90º diffuser
6. Light paint glass and metal objects with scrim lighting or use Savage Edge Lit Mini diffuser light
7. Light top surfaces from above with 160 LED panel or light saber
8. Remove black background material and light paint background, making sure NOT to light any of your subject pieces. Place the light in front of you at at just below belly height pointing toward the backrgound and walk from one side of the scene to the other (your body will illuminate the background while preventing light from hitting your subject

Light Painting Steps–Larger Subjects

1. Once composition, view angle and focal length are determined, finalize focus using manual focus mode
2. For sunsets, capture sun as it descends below horizon; dusk is sometimes better
3. Be sure to take note of which lights you use on each shot (for future white balancing)
4. Create Fill Light layer by exposing ambient light after sunset just enough to provide detail in the areas not light by artificial lights (i.e. foreground grass or pavement, background grass or trees, etc.). Be sure to shoot a gray card or Color Checler Passport target for white balancing this shot later!

OR

4. Create Fill Light layer moving bright light behind, above and to the sides of the camera
5. Light paint larger objects first with LED panel or light saber
6. Light paint backlit areas with smaller lights
7. Light paint all background objects using large LED panel or spotlight, if possible (if not, see #5 above)
8. Light paint the ground with light at low angle for added texture
9. For vehicles, light the interior as well (open all windows before beginning

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Light Painting Steps–RAW Processing

IMPORT YOUR FILES

Bring your files into your RAW processor

WHITE BALANCE (videos #5, 6 and 7)

White balance each file with the custom white balance and/or profile presets you made for each of your light sources. Refer to your notes when shooting to make sure you assign the correct preset to each image (you *did* take notes, didn't you?)

ADJUST EXPOSURES

Use the Highlight slider to make minor exposure adjustments to each image. Use the Exposure slider only if major exposure changes (1 or more stops) are necessary. You should not be making any color or density adjustments in Photoshop, so make sure each photo is perfect in your RAW processor.

SHARPEN FILES (video #15)

By sharpening the RAW files, your images will be sharp regardless of their finished size.

EXPORT (video #12)

There are two ways to create your light painting Master File. One is to export all the files as full size #10 JPEGs (at 100% quality) into a separate folder, where they can be viewed and opened from Adobe Bridge. This is the method that is used in all the processing videos in the tutorial.

The other method is to export just the Base Layer image as a full size #10 JPEG (at 100% quality), opening and saving it at your Master File. Then from Lightroom or Adobe Camera Raw, open the other files into Photoshop one at a time, selecting the area you wish to use in each and playing the Set Up for Brushing Action. The RAW file will close and allow you to brush in the new layer on your Master File. Depending on the speed of your computer, it might take a little longer to open each RAW file into Photoshop, but you won't have to spend the time processing your RAW files into JPEGs that fill up your hard drive and won't ever be used again.

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Light Painting Steps–Photoshop

BASE LAYER

1. Open Base Layer image
2. Play "Make Base Layer" action
3. Image will appear black; that's OK, it's supposed to be
4. Save as Master File in .psb (Large Document) format

LIGHT PAINTING (video #8)

1. Use the Lasso tool (0 px feather) to select desired area, leaving room around the area you will be brushing in
2. Play "Set Up for Brushing" action
3. With Brush tool set to soft, Opacity set to 100% and Flow set to about 15%, and background set to white, begin brushing the area. This allows you to build up density by holding the pen down and brushing
4. If you brush too much back, hit the X key to change brush to black and paint in to reduce brightness
5. Shift-click on Layer Mask disables mask to see what the layer looks like (Shift-click again to enable mask)
6. Be sure to brush all specular highlights back to pure white—anything less will be a 'graylight'

ADDING THE BACKGROUND FOR STILL LIFE LIGHT PAINTINGS (video #9)

1. Open background photo and select all (Command/Control-A) and play "Set Up for Brushing" action
2. Brush in desired amount of background
3. With the new background layer selected, play the "Adjust Background" action, the double-click on the Hue Adjustment Layer to bring the Hue/Saturation properties window up
4. Adjust color or saturation of background to taste, independently of the foreground objects

HARVESTING DETAIL FROM THE BASE LAYER (video #11)

1. Click on the white Base Layer Mask
2. Click the "B" key to set Foreground and Background to black and white, then click the "X" key to change brush to black to blend transitions or add detail

MAKE FINAL EXPOSURE ADJUSTMENTS (video #14)

1. Place all layers into a layer Group (click in upper right corner of Layers palette and go to "New Group from Layers")
2. Create Levels Adjustment Layer
3. Move Input slider (on top) to the left until it almost reaches pixel data (or to taste)

REMOVE SENSOR DUST AND UNWANTED SPOTS AND HIGHLIGHTS (video #10)

1. Flatten file and save as a #10 quality Master File.jpg
2. Play the Nolan Effect action and zoom to 100% or 200% magnification
3. Carefully scroll through the file, removing spots with the Patch Tool or Healing Brush Tool

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Light Painting Keyboard Shortcuts

- Switch between Bridge and Photoshop: **Command-Tab (Mac) or Control-Tab (Windows)**

- Lasso tool: **L**

- Add to a selection: **Hold Shift key down while drawing**

- Subtract from a selection: **Hold Option key down while drawing**

- Brush tool: **B**

- Increase Brush size: **Click and hold the right Bracket key } or]**

- Increase Brush size: **Click and hold the left Bracket key { or [**

- Pen tool: **P**

- Zoom tool: **Z, then drag pen diagonally**

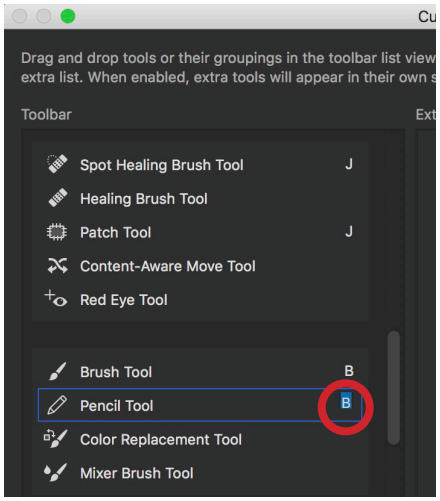
- Change Brush opacity: **B** (if Brush tool isn't selected)**, then click a number (7=70%, 25=25%, etc.)**

- Change Flow opacity: **B,** (if Brush tool isn't selected) **then shift-click a number (5=50%, 13=13%, etc.)**

- Find a specific layer: **V,** (if Move tool isn't selected) **then Control-click on the area. Flyout menu will show which layers are contained in the clicked spot. Select one of those layers in the menu and that layer will be selected in the Layers palette.** (This is a **VERY** valuable tool when trying to find a specific layer to edit!)

- Move around magnified window: **Hold spacebar and drag pen**

- **NOTE:** Many of the toolbar icons have similar tools nested within them, for example, the Brush, Pencil, Color Replacement and Mixer brush are all contained within the Brush icon. Hitting the B key will reveal the last icon you used, and hitting the B key repeatedly will cycle through all the available brushes. If you only use one or two tools most of the time, you can prevent them from appearing when you hit the keyboard shortcut. Go to Edit > Toolbar and click on the letter to the right of each tool you don't want and hit the Delete key. This saves you time and the annoyance of getting a tool you don't want. You still have full access to the tools in the toolbar, they just won't appear when you use the keyboard shortcut.



Select and delete the letter of the tool you don't want.

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